Travya: Agentic AI-Powered Travel Companion

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Abstract

Travya is an agentic AI-powered travel companion application designed to revolutionize travel planning by leveraging autonomous AI agents capable of reasoning, planning, adapting, and acting on behalf of users. Built with a fullstack FastAPI backend, React frontend, and Docker-based deployment, Travya integrates Google's Agent Development Kit (ADK) for multi-agent orchestration, Gemini models for reasoning, and Vertex AI for scalable AI operations. The system supports multi-modal inputs (text, image, voice), personalized itinerary planning via retrieval-augmented generation (RAG), real-time adaptations, and booking/payment integrations. Its architecture emphasizes modularity, loose coupling, high reliability (99.99% uptime), and cost efficiency ($500–2,000/month on GCP). Travya exemplifies a production-ready, scalable, and privacy-focused AI travel companion capable of delivering personalized, adaptive travel plans efficiently and reliably.

1 Literature Review

1.1 Introduction

The integration of artificial intelligence (AI) into the tourism and travel industry has transformed traditional planning processes into dynamic, personalized experiences [1]. Agentic AI, characterized by autonomous agents capable of reasoning, planning, adapting, and executing tasks, represents a significant advancement in this domain [2]. Unlike passive AI tools that respond to queries, agentic systems employ multi-agent collaboration to manage complex workflows, such as itinerary generation, booking, and real-time adjustments [3]. This literature review synthesizes existing research, platforms, and technologies relevant to Travya, an agentic AI-powered travel companion. Drawing from academic papers, industry reports, and real-world applications, it highlights trends, opportunities, challenges, and gaps. The review spans publications from 2019 to 2025, based on a comprehensive search of scholarly databases, web resources, and social media discussions conducted as of September 18, 2025.

Key themes include the shift toward personalized, multi-modal travel planning, the role of large language models (LLMs) like Gemini in enhancing user interactions, and the application of multi-agent systems for operational efficiency [4]. AI's potential to support sustainable tourism, such as through eco-friendly itinerary recommendations, is also emphasized [5]. However, challenges such as data privacy, ethical AI use, and integration with legacy systems persist [3].

1.2 Existing Platforms and Industry Applications

Several AI-powered travel companions have emerged, competing in a market projected to grow significantly due to post-pandemic demand for seamless planning [6]. Layla AI (formerly Roam Around) is a prominent example, offering tailored itineraries, road trip suggestions, and end-to-end planning via chat interfaces [7]. It supports multilingual interactions and integrates with booking systems, making it accessible for global users. GuideGeek, powered by Matador Network, focuses on real-time information for flights, accommodations, and experiences, enabling complex trip planning in minutes [8]. Mindtrip stands out for its map-based visualizations and semantic search, allowing users to query trips with natural language and receive structured outputs like visual itineraries [9].

Other platforms, such as Wonderplan, iPlan.ai, and Vacay, emphasize personalization through user preferences like budget and interests [10]. For instance, Booking.com leverages OpenAI's LLMs for smarter searches and intent-driven experiences, integrating data systems for real-time support [11]. IHG Hotels & Resorts uses Vertex AI and Gemini models to build travel planners that personalize recommendations [12]. Emerging agentic platforms like Flyte AI automate flight and hotel bookings with live agents [13], while Amadeus by AIREV predicts delays and tracks schedules using AI agents [14].

Social media discussions highlight practical implementations, such as MakeMyTrip's Myra, an agentic AI for multilingual trip planning [15], and Fetch.ai's ASI:One, which uses uAgents for itinerary creation based on user interests [16]. Travel Sage on Lilypad Network and Sentient's Travel Explorer employ multi-agent workflows for comprehensive planning [17, 18]. These platforms often incorporate retrieval-augmented generation (RAG) for personalization and tools like Google Places API for real-time data [19]. Critiques note limitations, such as over-reliance on AI potentially reducing user agency in planning [20].

1.3 Academic Papers and Research

Scholarly literature underscores AI's role in enhancing tourism efficiency and personalization. A systematic review by Gössling et al. [3] analyzes machine learning (ML) applications in tourism, identifying clusters in demand forecasting, recommender systems, and sentiment analysis, while noting data scarcity in underrepresented regions. Beyond tourism, Olujimi et al. [22] provide a systematic literature review of agentic AI in SMMEs, emphasizing ecosystemic interconnected agents that offer useful parallels for travel applications.

Key papers on agentic AI include Singh et al. [21], who introduced combining LLMs with automated planners for guaranteed travel itineraries. Borrajo et al. [24] extend this to explainable AI planning, addressing user trust in agentic systems. Gundawar et al. [23] propose robust LLM-modulo frameworks for travel, emphasizing resilience to uncertainties like weather.

On multi-agent systems, Borrajo et al. [24] discuss lazy incremental learning for quality plans, applicable to agent orchestration. Recent works by Arora et al. [25] explore AI-driven personalization in Uzbekistan, demonstrating improved tourist experiences via Gemini-

Table 1: Comparison of AI-Powered Travel Platforms

Platform	Key Features	Strengths	Limitations
Layla AI	Chat-to-booking, multilingual, maps	Personalized itineraries, ease of use	Limited integration with niche providers
GuideGeek	Real-time flight/stay info	Quick complex planning	Primarily English-based
Mindtrip	Semantic search, visual outputs	Multi-modal queries	Higher cost for premium features
Myra (MakeMyTrip)	Agentic multilingual planning	Real-time decision handling	Region-specific focus (e.g., India)
ASI:One (Fetch.ai)	uAgents for itineraries	Blockchain-integrated, decentralized	Requires Web3 familiarity

like models. Anitha [26] details an AI platform for itinerary generation, mirroring Travya's modularity.

Sustainability-focused research, such as Siddik et al. [5], examines AI's role in economic cycles, linking adoption to eco-tourism growth using data from 2010–2022. Seyfi et al. [27] investigate barriers to generative AI in decision-making, moderated by personality traits. Husáková [28] explore generational differences in AI adoption for smart tourism.

Overviews by Buhalis et al. [1] and Hassannia et al. [29] synthesize AI's transformative impact, noting contributions from robotics and big data. López-Naranjo et al. [30] integrate practitioner and expert views, predicting trends in AI for marketing and operations.

1.4 Key Technologies and Frameworks

Google's Agent Development Kit (ADK) facilitates multi-agent systems, optimized for Gemini and Vertex AI, enabling flexible agent deployment [4]. Gemini models, via Vertex AI, support multimodal inputs for travel apps, as demonstrated in Google's agent-powered planning tutorial [31]. Multi-agent frameworks like CrewAI and Amazon Nova balance cost and performance for workflows [32]. Observability tools ensure reliability in agentic systems [2].

1.5 Trends, Challenges, and Gaps

Trends include autonomous agents for dynamic experiences and AI's role in sustainability [33]. Challenges encompass geographical biases, diverse preferences, and ethical risks like data privacy [3]. Gaps exist in fully autonomous, GDPR-compliant multi-agent systems for eco-trips and AR previews—areas where Travya innovates by leveraging ADK, Gemini, and modular design for scalable, privacy-focused planning.

This review positions Travya as a competitive, production-ready solution addressing these gaps through agentic orchestration and real-time adaptations. Future research should explore long-term user adoption and AI's socioeconomic impacts in tourism [6].

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